

Week 3

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GNU nano 8.7.1 week-3.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    pid_t pid = fork();

    if (pid < 0) {
        printf("Fork failed\n");
        exit(1);
    } else if (pid == 0) {
        // Child process
        printf("Child Process:\n");
        printf("PID: %d, PPID: %d\n", getpid(), getppid());
        sleep(2); // simulate waiting
        printf("Child finished execution\n");
        exit(0);
    } else {
        // Parent process
        printf("Parent Process:\n");
        printf("PID: %d, Child PID: %d\n", getpid(), pid);
        wait(NULL); // parent waits for child
        printf("Child terminated, parent continues\n");
    }
    return 0;
}
```

```
adith@KesavaSatish:~$ gcc week-3.c
```

```
adith@KesavaSatish:~$ ./a.out
```

```
Parent Process:
```

```
Child Process:
```

```
PID: 633, PPID: 632
```

```
PID: 632, Child PID: 633
```

```
Child finished execution
```

```
Child terminated, parent continues
```